GD file (created by the user)

Load GD file in memory

String

Generated by ANTLR4
(from the GD-language grammar)

Lexer (lexical analysis)

List of Tokens

Generated by ANTLR4
(from the GD-language grammar)

Parser (syntax analysis)

Parse Tree

Type Checker

Parse-Tree Visitor

Generated by ANTLR4 and
Extended (s.t. it generates the AST upon tree traversal)

Abstract Syntax Tree (AST)

Abstract Syntax Tree (AST)

AST transformations (optimization)

AST (simplified & optimized)

BackEnd AST-Optimizer (CPU-Serial-C)

AST (backend-optimized)

Code-Generator (CPU-Serial-C)

C file

C compiler

binary file

execute (on single-core)

BackEnd AST-Optimizer (GPU-Parallel-C-OpenCL)

AST (backend-optimized)

Code-Generator (GPU-Parallel-C-OpenCL)

C file (host code)

C compiler

binary file

OpenCL kernel

OpenCL library
(with kernel compiler)

execute (on GPU-equipped machine)